CSGE602055 Operating Systems
CSF2600505 Sistem Operasi
Week 01: Overview 2, Virtualization & Scripting

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https://os.vlsm.org/Slides/os00.pdf
Always check for the latest revision!

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OS212[4]): Operating Systems 2021 - 2

OS A	OS B	OS C	OS INT
Every first day of the Week, **Quiz#1:** (07:40-07:50) and **Quiz#2:** 07:20-07:40			
Monday/Thursday 13:00 — 14:40 14:00 — finish	Monday/Thursday 15:00 — 16:40[1] 16:00 — finish	Monday/Thursday 13:00 — 14:40 13:00 — 14:40	Monday/Wednesday 08:00 — 09:40 09:00 — finish

Week	**Schedule & Deadline**[2])	**Topic**	**OSC10**[3])
Week 00	30 Aug - 05 Sep 2021	Overview 1, Virtualization & Scripting	Ch. 1, 2, 18.
Week 01	06 Sep - 12 Sep 2021	Overview 2, Virtualization & Scripting	Ch. 1, 2, 18.
Week 02	13 Sep - 19 Sep 2021	Security, Protection, Privacy, & C-language.	Ch. 16, 17.
Week 03	20 Sep - 26 Sep 2021	File System & FUSE	Ch. 13, 14, 15.
Week 04	27 Sep - 03 Oct 2021	Addressing, Shared Lib, & Pointer	Ch. 9.
Week 05	04 Oct - 10 Oct 2021	Virtual Memory	Ch. 10.
Week 06	11 Oct - 31 Oct 2021	Concurrency: Processes & Threads	Ch. 3, 4.
Week 07	01 Nov - 07 Nov 2021	Synchronization & Deadlock	Ch. 6, 7, 8.
Week 08	08 Nov - 14 Nov 2021	Scheduling + W06/W07	Ch. 5.
Week 09	15 Nov - 21 Nov 2021	Storage, Firmware, Bootloader, & Systemd	Ch. 11.
Week 10	22 Nov - 28 Nov 2021	I/O & Programming	Ch. 12.

[1]) **OS B:** Week00-Week05 (RMS); Week06-Week10 (MAM).

[2]) The **DEADLINE** of Week 00 is 05 Sep 2021, whereas the **DEADLINE** of Week 01 is 12 Sep 2021, and so on...

[3]) Silberschatz et. al.: **Operating System Concepts**, 10th Edition, 2018.

[4]) This information will be on **EVERY** page two (2) of this course material.

STARTING POINT — https://os.vlsm.org/

- **Text Book** — Any recent/decent OS book. Eg. (**OSC10**) Silberschatz et. al.: **Operating System Concepts**, 10th Edition, 2018. See also https://www.os-book.com/OS10/.
- **Resources**
  - **SCELE OS212** — https://scele.cs.ui.ac.id/course/view.php?id=3268. The enrollment key is **XXX**.
  - **Download Slides and Demos from GitHub.com** https://github.com/UI-FASILKOM-OS/SistemOperasi/: os00.pdf (W00), os01.pdf (W01), os02.pdf (W02), os03.pdf (W03), os04.pdf (W04), os05.pdf (W05), os06.pdf (W06), os07.pdf (W07), os08.pdf (W08), os09.pdf (W09), os10.pdf (W10).
  - **Problems** 195.pdf (W00), 196.pdf (W01), 197.pdf (W02), 198.pdf (W03), 199.pdf (W04), 200.pdf (W05), 201.pdf (W06), 202.pdf (W07), 203.pdf (W08), 204.pdf (W09), 205.pdf (W10).
  - **LFS** — http://www.linuxfromscratch.org/lfs/view/stable/
  - **OSP4DISS** — https://osp4diss.vlsm.org/
  - **DOIT** — https://doit.vlsm.org/001.html

Agenda

1. Start
2. Schedule
3. Agenda
4. Week 01
5. Week 01: Review 2

Agenda (2)

Week 01 Overview II: Topics[1]

- Intelectual Property Rights (IPR)
- Software Licenses and Free Software
- Operating System Services and Interfaces
- System Calls and System Programming
- Types of virtualization (including Hardware/Software, OS, Server, Service, Network)
- Hypervisors
- Portable and cost of virtualization; emulation vs. isolation
- Cloud services: IAAS, PAAS and Platform APIs, SAAS
- Introduction to Scripting and REGEX.

[1] Source: ACM IEEE CS Curricula 2013

Week 01 Overview II: Learning Outcomes[1]

- Explain the concept of virtual memory and how it is realized in hardware and software. [Familiarity]
- Discuss hypervisors and the need for them in conjunction with different types of hypervisors. [Usage]
- Differentiate emulation and isolation. [Familiarity]
- Evaluate virtualization trade-offs. [Assessment]
- Discuss the importance of elasticity and resource management in cloud computing. [Familiarity]
- Explain the advantages and disadvantages of using the virtualized infrastructure. [Familiarity]

[1] Source: ACM IEEE CS Curricula 2013

The Operating System

- TBA